Workload validation of the generated seccomp profiles: all 20 containers

Each image runs a realistic workload in three modes, three times each: **base** (Docker's default seccomp profile), **log** (the generated profile with defaultAction SCMP_ACT_LOG: nothing is blocked, every syscall outside the profile is written to the audit log and attributed to the container's processes) and **enforce** (the generated profile as published). **PASS** = in every repetition, enforce completes the same operations as base with 0 failures, and log mode records no syscall outside the profile. A positive control (JRuby with mkdir and rename removed from its profile) reports exactly mkdir and rename, so silence in log mode is meaningful. Workloads are driven from the host or an unconfined client container; Groovy and JRuby are batch programs.

Result with the published profiles: 20 of 20 PASS (180 runs; 74,754 operations completed under the enforced profiles, 0 failed; no syscall outside any profile in the 60 log runs). The profiles are the ones in the paper (after the 2026-11-11 analysis fixes, which this validation motivated; see section 3).

1. JDK 8 set (Debian 10, Temurin 8u504; paper Table: SysJava vs SWAT4J)

App	Image	Profile	Workload	Ops / run	Base ok / failed (3 runs)	Log ok / failed (3 runs)	Enforce ok / failed (3 runs)	Outside profile (log)	Result
ZooKeeper	zookeeper:3.4.14-jdk8-buster	192	four-letter commands; 3 zkCli rounds of ~1,300 creates/sets/gets/deletes	3,951	11,853 / 0	11,853 / 0	11,853 / 0	none	PASS
Tomcat	tomcat:7.0-jdk8-buster	173	3,000 GETs on servlets/JSPs/docs, 50 POSTs, war hot deploy + undeploy	3,052	9,156 / 0	9,156 / 0	9,156 / 0	none	PASS
Cassandra	cassandra:3.0.29-jdk8-buster	230	cqlsh DDL/DML; cassandra-stress write/read/mixed, n=60,000 each (counted per step)	11	33 / 0	33 / 0	33 / 0	none	PASS
Jetty	jetty:9.4.51-jdk8-buster	161	3,000 GETs on static files of a deployed war, 50 POSTs, war hot deploy + undeploy	3,052	9,156 / 0	9,156 / 0	9,156 / 0	none	PASS
Storm	storm:2.4.0-jdk8-buster	179	nimbus confined; client runs storm list x10 and blobstore create/list/cat/update/delete x5	35	105 / 0	105 / 0	105 / 0	none	PASS
OrientDB	orientdb:3.2.19-jdk8-buster	174	REST: create db, classes, index, 1,500 inserts, edges, queries, update, drop db	1,808	5,424 / 0	5,424 / 0	5,424 / 0	none	PASS
Elasticsearch	elasticsearch:7.5.2-jdk8-buster	185	2 indices x 20 x 1,000-doc bulk, searches, aggregations, forcemerge, update_by_query, reindex, delete	250	750 / 0	750 / 0	750 / 0	none	PASS
Solr	solr:8.7.0-jdk8-buster	201	create core, 5 indexing rounds, 120 queries, optimize, backup, unload with delete	199	597 / 0	597 / 0	597 / 0	none	PASS
Groovy	groovy:4.0.0-jdk8-buster	189	batch script: files, regex, rename, JSON, XML, zip, threads, local HTTP server, Java interop	46	138 / 0	138 / 0	138 / 0	none	PASS
JRuby	jruby:9.4.2.0-jdk8-buster	168	batch script: files, regex, rename, chmod, JSON, zlib, threads, TCP server, Java interop	46	138 / 0	138 / 0	138 / 0	none	PASS

2. Latest set (newest image of each application; paper Table III)

App	Image	Profile	Workload	Ops / run	Base ok / failed (3 runs)	Log ok / failed (3 runs)	Enforce ok / failed (3 runs)	Outside profile (log)	Result
ZooKeeper	zookeeper:3.9.5 (JDK 17)	196	four-letter commands; 3 zkCli rounds of ~1,300 commands (deleteall instead of rmr)	3,951	11,853 / 0	11,853 / 0	11,853 / 0	none	PASS

App	Image	Profile	Workload	Ops / run	Base ok / failed (3 runs)	Log ok / failed (3 runs)	Enforce ok / failed (3 runs)	Outside profile (log)	Result
Tomcat	tomcat:11.0.22 (JDK 25)	180	3,000 GETs on servlets/JSPs/docs, 50 POSTs, war hot deploy + undeploy (examples without the localhost-only valve)	3,052	9,156 / 0	9,156 / 0	9,156 / 0	none	PASS
Cassandra	cassandra:5.0.8 (JDK 17)	233	cqlsh DDL/DML; cassandra-stress write/read/mixed, n=60,000 each (counted per step)	11	33 / 0	33 / 0	33 / 0	none	PASS
Jetty	jetty:12.1.10 (JDK 25)	191	bare server as shipped (no webapps): 3,000 GETs, 50 POSTs, HEAD/OPTIONS/PUT/DELETE, 20 keep-alive connections x 50 requests, an oversized header	3,075	9,225 / 0	9,225 / 0	9,225 / 0	none	PASS
Storm	storm:2.8.8 (JDK 17)	207	nimbus confined; client runs storm list x10 and blobstore create/list/cat/update/delete x5	35	105 / 0	105 / 0	105 / 0	none	PASS
OrientDB	orientdb:3.2.53 (JDK 8)	183	REST: create db, classes, index, 1,500 inserts, edges, queries, update, drop db	1,808	5,424 / 0	5,424 / 0	5,424 / 0	none	PASS
Elasticsearch	elasticsearch:7.1.0 (JDK 12)	208	2 indices x 20 x 1,000-doc bulk, searches, aggregations, forcemerge, update_by_query, reindex, delete	250	750 / 0	750 / 0	750 / 0	none	PASS
Solr	solr:10.0.0 (JDK 25)	224	SolrCloud (Solr 10 default): create collection, 5 indexing rounds, 120 queries, optimize, backup, delete collection	194	582 / 0	582 / 0	582 / 0	none	PASS
Groovy	groovy:5.0.6 (JDK 21)	191	batch script: files, regex, rename, JSON, XML, zip, threads, local HTTP server, Java interop	46	138 / 0	138 / 0	138 / 0	none	PASS
JRuby	jruby:10.1.0.0 (JDK 21)	187	batch script: files, regex, rename, chmod, JSON, zlib, threads, TCP server, Java interop	46	138 / 0	138 / 0	138 / 0	none	PASS

3. Before the fixes: what the validation found

The first round (same harness, the profiles before 2026-11-11) completed every operation in base and log for all 20 images. Log mode found syscalls missing from the profiles, and Tomcat 11 failed under enforce. Each finding was traced to its source and fixed in the analysis (not by adding syscalls by hand): libjvm's internals via its imports, vtable functions as start functions of C++ libraries, the runc whitelist (118 -> 123 entries), and for the latest set the C-library functions looked up with dlsym. The profiles grew by the sizes shown; the second round is sections 1 and 2.

Set	App	Profile before -> after	Enforce failures before	Outside profile before (log)	Source -> fix
JDK 8	Tomcat	170 -> 173	0	fchdir	fchdir: libjvm (hsperfdata) -> libjvm imports
JDK 8	Jetty	157 -> 161	0	fchdir, fchown	fchdir: libjvm (hsperfdata) -> libjvm imports; fchown: runc -> runc whitelist
JDK 8	Storm	172 -> 179	0	fallocate, fchdir	fallocate: RocksDB virtual call -> vtable starts; fchdir: libjvm (hsperfdata) -> libjvm imports
JDK 8	Elasticsearch	183 -> 185	0	fchown	fchown: runc -> runc whitelist
JDK 8	Solr	199 -> 201	0	times	times: libjvm (G1 pause timing) -> libjvm imports
latest	ZooKeeper	191 -> 196	0	openat2	openat2: runc 1.3.4 -> runc whitelist
latest	Tomcat	174 -> 180	250 of 3,052 per run	copy_file_range, flock, openat2	copy_file_range: JDK 25 libnio, looked up with dlsym -> dlsym rule; flock: libjvm (JDK 21/25) -> libjvm imports; openat2: runc 1.3.4 -> runc whitelist
latest	Cassandra	230 -> 233	0	openat2	openat2: runc 1.3.4 -> runc whitelist

Set	App	Profile before -> after	Enforce failures before	Outside profile before (log)	Source -> fix
latest	Jetty	186 -> 191	0	openat2	openat2: runc 1.3.4 -> runc whitelist
latest	Storm	202 -> 207	0	openat2, sync_file_range	openat2: runc 1.3.4 -> runc whitelist; sync_file_range: RocksDB 8 virtual call -> vtable starts
latest	OrientDB	175 -> 183	0	close_range, openat2	close_range: runc 1.3.4 -> runc whitelist; openat2: runc 1.3.4 -> runc whitelist
latest	Elasticsearch	199 -> 208	0	close_range, faccessat2, openat2	close_range: runc 1.3.4 -> runc whitelist; faccessat2: runc 1.3.4 -> runc whitelist; openat2: runc 1.3.4 -> runc whitelist
latest	Solr	221 -> 224	0	openat2	openat2: runc 1.3.4 -> runc whitelist
latest	Groovy	187 -> 191	0	openat2	openat2: runc 1.3.4 -> runc whitelist
latest	JRuby	182 -> 187	0	flock, openat2	flock: libjvm (JDK 21/25) -> libjvm imports; openat2: runc 1.3.4 -> runc whitelist

Tomcat 11's 250 failures per run are GET /docs/config/http.html (a large static file): JDK 25's libnio finds copy_file_range with dlsym and calls it through a pointer, so no call graph shows it. The JDK 8 Storm profile in the first round was the 172-syscall one; Solr's rmdir (needed for core unload) came from RTA, before this validation.

4. Where the data is

```
JDK 8:  ~/Tutorial/EchoTrace/workload_validation/       REPRODUCE.md, run_validation.py, make_table.py
        results/ (before the fixes), results_fixes/ (published profiles), control/ (positive control)
        Workload_validation_and_binding_JDK8.pdf (JDK 8 details + binding ablation)
latest: ~/Tutorial/EchoTrace_latest/workload_validation/ REPRODUCE.md, run_validation.py, make_table.py
        results_current/ (before the fixes), results_fixes/ (published profiles), control/
This PDF: python make_pdf_all20.py (in the JDK 8 folder; needs reportlab)
```